

Advanced Statistics & probability

Lecture 1

By

Dr/ Mohamed E. Sobh

Contents of This Course:

- ❑ **Chapter (1):** Normal Distribution.
- ❑ **Chapter (2):** Law of Large Numbers and Central Limit Theorem .
- ❑ **Chapter (3):** Distributions Derived From Normal Distributions (Chi-Squared, Student-t, and F distribution).
- ❑ **Chapter (4):** Statistical Estimation, Point Estimation, Confidence Interval.
- ❑ **Chapter (5):** Test of Hypothesis, Fitting Straight Lines.
- ❑ **Chapter (6):** Analysis of Variance, Stochastics Models, Poisson Processes.

Methods of students' assessment:

- ❑ **Course Work: 10Marks.**
- ❑ **Project: 20Marks.**
- ❑ **Midterm Exam: 20Marks.**
- ❑ **Final Exam: 50Marks.**
- ❑ **Total: 100Marks.**

Teaching and Learning Methods :

- ☐ **Lectures.**
- ☐ **Tutorial.**
- ☐ **Self-learning.**
- ☐ **Solving problem.**
- ☐ **Discussion.**

Chapter (1)

Normal Distribution

- ❑ One of the most important continuous distribution.
- ❑ Many measurable characteristics are normally distributed or approximately normally distributed.
- ❑ The normal depend on two parameters which are mean (μ) and variance (σ^2).
- ❑ The probability density function of the normal distribution is given in the form:

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

□ The continuous random variable **X** which has a normal distribution has several important characteristics:

- 1) $-\infty < x < \infty$.
- 2) The density function of X , $f(x)$ has a bell-shaped curve and never touches the x -axis.
- 3) The area under the normal curve equals one.
- 4) The highest point of the curve of $f(x)$ is at the mean (μ).
- 5) The curve of $f(x)$ is symmetric about the mean (μ) (i.e., mean=median=mode).
- 6) Normal distributions are denser in the center and less dense in the tails.
- 7) If the random variable **X** is normally distributed with mean (μ) and variance (σ^2), we write

$$X \sim \text{Normal}(\mu, \sigma^2) \quad \text{or} \quad X \sim N(\mu, \sigma^2)$$

8) We note that about **68%** of area of a normal distribution is within **one** standard deviation of the mean

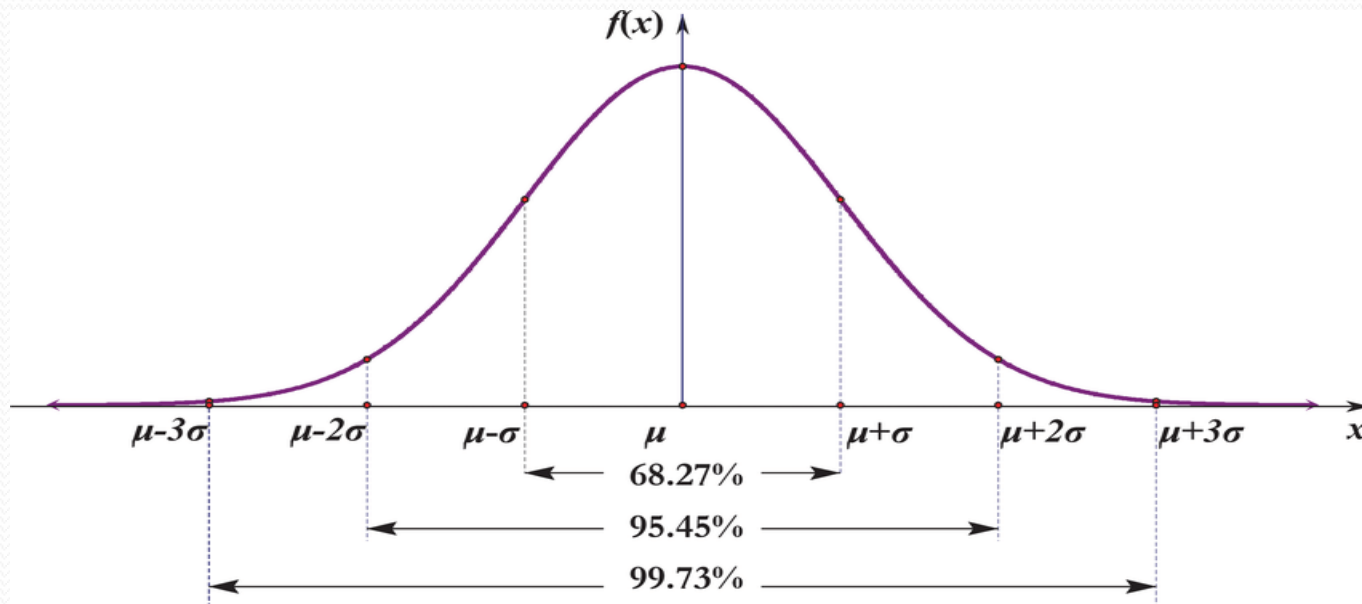
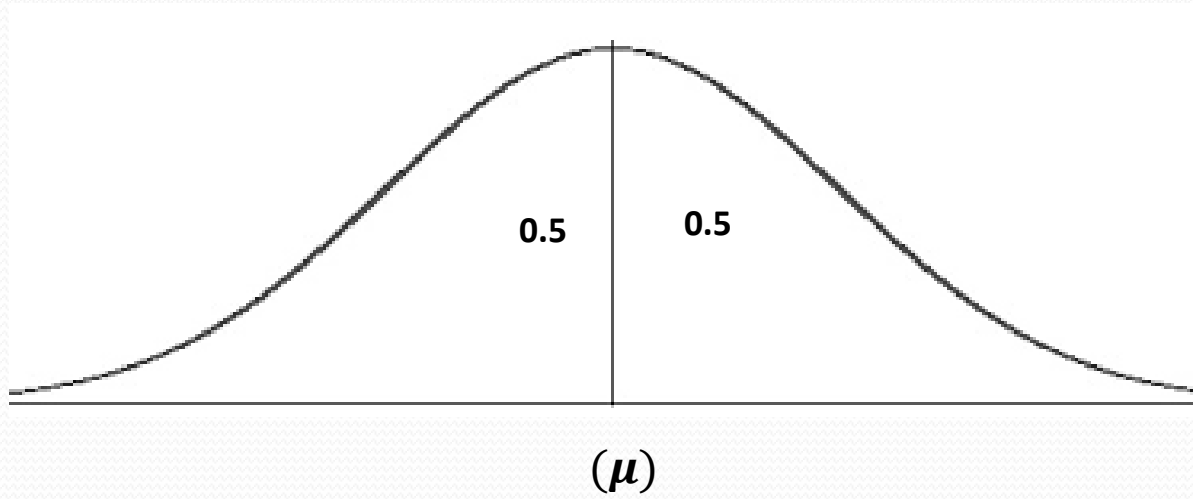
$$P(\mu - \sigma < X < \mu + \sigma) = 68\%$$

9) Approximately **95%** of area of a normal distribution is within **two** standard deviation of the mean

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 95\%$$

10) Approximately **99.7%** of area of a normal distribution is within **three** standard deviation of the mean

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 99.7\%$$



Standard Normal Distribution

❑ The standard normal distribution is very important because the probabilities of any normal distribution is calculated from the probabilities of standard normal distribution

❑ The normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$ is called standard normal distribution and is denoted by $Z \sim N(0, 1)$.

❑ The continuous random variable X which has a normal distribution has several important characteristics:

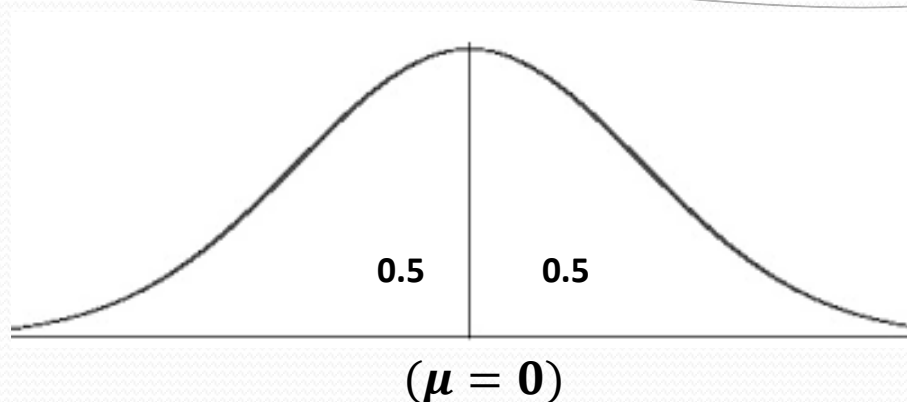
1) $-\infty < z < \infty$.

2) The density function of Z has a bell-shaped curve and never touch x -axis.

3) The area under normal curve equal one.

4) The highest point of the curve of standard normal distribution at $\mu = 0$.

5) The curve of standard normal is symmetric about $\mu = 0$ (i.e., mean=median=mode).



❑ In order to calculate the probabilities of a random variable $X \sim N(\mu, \sigma^2)$ we must convert to standard normal distribution (Change X to Z).

❑ Changing from normal to standard normal:

$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

□ **Remark:**

$$i. P(X < a) = P\left(\frac{X-\mu}{\sigma} < \frac{a-\mu}{\sigma}\right) = P\left(Z < \frac{a-\mu}{\sigma}\right) = \varphi\left(\frac{a-\mu}{\sigma}\right).$$

$$ii. P(X > a) = P\left(\frac{X-\mu}{\sigma} > \frac{a-\mu}{\sigma}\right) = P\left(Z > \frac{a-\mu}{\sigma}\right) = 1 - \varphi\left(\frac{a-\mu}{\sigma}\right)$$

$$iii. P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right) = P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) = \varphi\left(\frac{b-\mu}{\sigma}\right) - \varphi\left(\frac{a-\mu}{\sigma}\right).$$

$$iv. \varphi(-a) = 1 - \varphi(a).$$

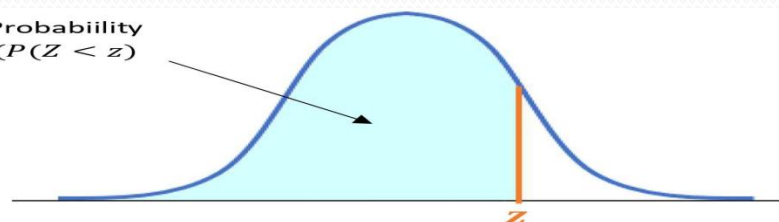
$$v. P(Z < 0) = P(Z > 0) = 0.5$$

$$vi. \varphi(0) = 0.5$$

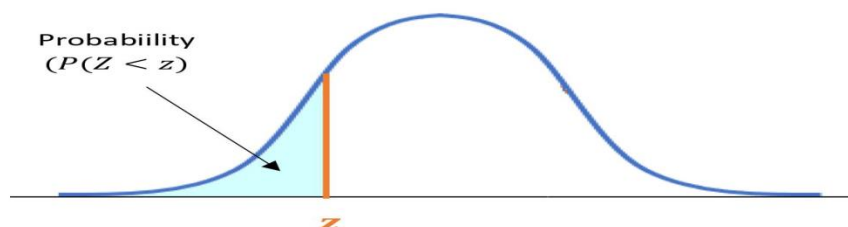
The value of $\varphi(\cdot)$ are obtained from standard normal table .



Probability
($P(Z < z)$)



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9700	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9762	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9983	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.5000	0.4960	0.4920	0.4880	0.4841	0.4801	0.4761	0.4721	0.4681	0.4641
-0.10	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.20	0.4207	0.4168	0.4129	0.4091	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.30	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.40	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.50	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.60	0.2743	0.2709	0.2676	0.2644	0.2611	0.2579	0.2546	0.2514	0.2483	0.2451
-0.70	0.2420	0.2389	0.2358	0.2327	0.2297	0.2266	0.2236	0.2207	0.2177	0.2148
-0.80	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.90	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1.00	0.1587	0.1563	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.10	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.20	0.1151	0.1131	0.1112	0.1094	0.1075	0.1057	0.1038	0.1020	0.1003	0.0985
-1.30	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.40	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
-1.50	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.60	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.70	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.80	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.90	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2.00	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.10	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.20	0.0139	0.0136	0.0132	0.0129	0.0126	0.0122	0.0119	0.0116	0.0113	0.0110
-2.30	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.40	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0070	0.0068	0.0066	0.0064
-2.50	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.60	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.70	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.80	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.90	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3.00	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-3.10	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.20	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.30	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.40	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002	0.0002

- Example:

If X is a normally distributed random variable represent the hemoglobin level for healthy adults male with mean **16** and variance **0.81**. find the probability that healthy adult male has hemoglobin level less than or equal **14**.

Solution

$$\mu = 16,$$

$$\sigma^2 = 0.81,$$

$$\sigma = 0.9$$

$$\begin{aligned} P(X \leq 14) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{14 - 16}{0.9}\right) = P(Z \leq -2.22) \\ &= \varphi(-2.22) = 0.0132 \end{aligned}$$

- Example:

The weight of students of certain school are normally distributed with mean **40** kg and standard deviation **5** kg. Find the percentage of students whose weight

- a) Less than **50** kg.
- b) Between **30** kg and **55** kg.
- c) More than **45** kg.

Solution

$$\mu = 40, \quad \sigma = 5$$

$$a) P(X < 50) = P\left(\frac{X-\mu}{\sigma} < \frac{50-40}{5}\right) = P(Z < 2) = \varphi(2) = 0.9773.$$

$$b) P(30 < X < 55) = P\left(\frac{30-40}{5} < \frac{X-\mu}{\sigma} < \frac{55-40}{5}\right) = P(-2 < Z < 3) = \varphi(3) - \varphi(-2) = 0.9987 - 0.0228 = 0.9759.$$

$$c) P(X > 45) = P\left(\frac{X-\mu}{\sigma} > \frac{45-40}{5}\right) = P(Z > 1) = 1 - \varphi(1) = 1 - 0.8413 = 0.1587.$$

- Example:

A Certain type of storage battery lasts on the average **3** year, with standard deviation **0.5** year Assume that battery lives are normally distributed find the probability that battery will lasts less than **2.3** year.

Solution

$$\mu = 3, \quad \sigma = 0.5$$

$$P(X < 2.3) = P\left(\frac{X - \mu}{\sigma} < \frac{2.3 - 3}{0.5}\right) = P(Z < -1.4)$$

$$= \varphi(-1.4) = 1 - \varphi(1.4) = 0.0808$$

- Example:

It is known that the waiting time (in minute) of Patients at a certain clinic are $N(12, 16)$, find the Percentage of Patients who wait at clinic:

- a) For less than 9 minutes.
- b) For more than 10 minutes.
- c) Between 8 and 14 minutes.

Solution

$$\mu = 12, \quad \sigma^2 = 16, \quad \sigma = 4$$

$$a) P(X < 9) = P\left(\frac{X-\mu}{\sigma} < \frac{9-12}{4}\right) = P(Z < -0.75) = \varphi(-0.75) = 0.2266.$$

$$b) P(X > 10) = P\left(\frac{X-\mu}{\sigma} > \frac{10-12}{4}\right) = P(Z > -0.25) = 1 - \varphi(-0.25) = 1 - 0.4013 = 0.5987.$$

$$\begin{aligned} c) P(8 < X < 14) &= P\left(\frac{8-12}{4} < \frac{X-\mu}{\sigma} < \frac{14-12}{4}\right) = P(-1 < Z < -0.25) = \varphi(-0.25) - \varphi(-1) \\ &= 0.4013 - 0.1587 = 0.2426. \end{aligned}$$

- Example:

The height of **1000** student in a certain College are normally distributed with mean **68** inches and standard deviation of **3** inches. How many of these students would You expected to have heights:

- a) less than 64 inches.
- b) Between 67 and 71 inches.

Solution

$$\mu = 68, \quad \sigma = 3, \quad N = 1000$$

$$a) P(X < 64) = P\left(\frac{X - \mu}{\sigma} < \frac{64 - 68}{3}\right) = P(Z < -1.33) = \varphi(-1.33) = 0.0918.$$

$$\Rightarrow \text{Number of student} = 0.0918 * 1000 = 91.8 \approx 92$$

$$\begin{aligned} b) P(67 < X < 71) &= P\left(\frac{67 - 68}{3} < \frac{X - \mu}{\sigma} < \frac{71 - 68}{3}\right) = P(-0.33 < Z < 1) = \varphi(1) - \varphi(-0.33) \\ &= 0.8413 - 0.3707 = 0.4706. \end{aligned}$$

$$\Rightarrow \text{Number of student} = 0.4706 * 1000 = 470.6 \approx 471$$

- Example:

Let X has $N(8, \sigma^2)$ Find:

a) $P(8 - \sigma < X < 8 + \sigma)$.

b) $P(8 - 2\sigma < X < 8 + 3\sigma)$.

Solution

$$\mu = 8$$

$$a) \quad P(8 - \sigma < X < 8 + \sigma) = P\left(\frac{(8-\sigma)-8}{\sigma} < \frac{X-\mu}{\sigma} < \frac{(8+\sigma)-8}{\sigma}\right) = P(-1 < Z < 1) =$$

$$\varphi(1) - \varphi(-1) = 0.8413 - 0.1587 = 0.6826.$$

$$b) \quad P(8 - 2\sigma < X < 8 + 3\sigma) = P\left(\frac{(8-2\sigma)-8}{\sigma} < \frac{X-\mu}{\sigma} < \frac{(8+3\sigma)-8}{\sigma}\right) = P(-2 < Z < 3) =$$

$$\varphi(3) - \varphi(-2) = 0.9987 - 0.0228 = 0.9759.$$

• **Example:**

Let X has $N(12, 9)$ Find:

- a) The value of k such that $P(X < k) = 0.9875$.
- b) The value of k such that $P(X > k) = 0.25$.

Solution

$$\mu = 12, \quad \sigma^2 = 9, \quad \sigma = 3$$

$$a) \quad P(X < k) = P\left(\frac{X-\mu}{\sigma} < \frac{k-12}{3}\right) = P\left(Z < \frac{k-12}{3}\right) = \varphi\left(\frac{k-12}{3}\right) = 0.9875.$$

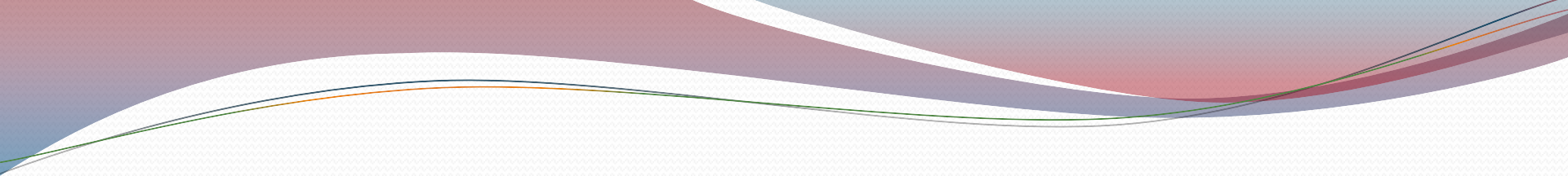
From standard normal table we have:

$$\frac{k-12}{3} = 2.24 \quad \Rightarrow \quad k-12 = 6.72 \quad \Rightarrow \quad k = 18.72$$

$$b) \quad P(X > k) = P\left(\frac{X-\mu}{\sigma} > \frac{k-12}{3}\right) = P\left(Z > \frac{k-12}{3}\right) = 1 - \varphi\left(\frac{k-12}{3}\right) = 0.25.$$

which implies $\varphi\left(\frac{k-12}{3}\right) = 0.75$. From standard normal table we have:

$$\frac{k-12}{3} = 0.675 \Rightarrow k-12 = 2.025 \Rightarrow k = 14.025$$



Thank you for your
Attention!

